

Non-Additive Strain–Doping Coupling in Quasi-Hexagonal C₆₀ Graphullerene Networks

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Materials design often treats strain and chemical substitution as independent control parameters. For quasi-hexagonal (qHP) C₆₀ graphullerene, we show that assumption is quantitatively wrong: (i) strain sensitivity $\alpha = dE/d\epsilon$ changes sign and magnitude with dopant type ($\alpha_{\text{pristine}} \approx -1$, $\alpha_N = -303$, $\alpha_B = +57$ meV/%; Exp. 5); (ii) the synergy order parameter $\mathcal{S}$ [Eq. (1)] reaches $|\mathcal{S}| \sim 7\text{--}24$ meV/atom at tetramer scale under +3% strain—far above the additive limit $\mathcal{S} = 0$; (iii) $|\mathcal{S}| > 0$ persists for $n = 1\text{--}8$ supercells (Exp. 10, 40/40 DFT points). These results define a coupled (ϵ, δ) design space for tuning stability of doped qHP networks rather than separate strain or doping scans.

I. INTRODUCTION

Graphullerenes—two-dimensional networks of qHP C₆₀ molecules—are tunable via substitution and mechanical load [1–4]. Prior work maps doping *or* strain separately; the *joint* response of total energy to (ϵ, δ) under identical protocols is unreported for controlled B/N/P substitution.

Gap. If $E(\epsilon, \delta)$ were additive, a two-scan workflow (strain on pristine + substitution at $\epsilon = 0$) would predict combined states. No metric in the literature tests that residual for qHP C₆₀ across dopants and sizes.

This work delivers three verified outcomes from PBE+DFT-D3 (CP2K): (1) opposite-sign α for N vs. B under the same biaxial grid; (2) non-zero $\mathcal{S}$ at +3% strain quantifying the additive-model error; (3) finite-size scaling to $8 \times \text{C}_{60}$ showing the cross term does not vanish at the largest cells computed. Section II defines $\mathcal{S}$; Sec. III reports numbers; Sec. IV states design implications and limits.

II. COMPUTATIONAL METHODS

Electronic structure calculations use CP2K [5] with PBE and Grimme DFT-D3 (Becke–Johnson damping), matching `experiments/*/inputs/*.inp`. Plane-wave cutoff is 300 Ry (280 Ry for $n \geq 6$ in Exp. 10); SCF convergence uses `EPS_SCF` = 10^{-6} with OT minimization. Basis: DZVP-MOLOPT-SR-GTH with matching GTH pseudopotentials.

Systems. Exp. 5: C₆₀ tetramers (240 atoms), six-fold B/N/P substitution, biaxial strain $\epsilon \in \{-5, -2.5, 0, 2.5, 5\}\%$ (rigid cell scaling, fixed fractional coordinates). Exp. 10: $n \times \text{C}_{60}$ ($n = 1\text{--}8$) at $\epsilon = 0$ and +3% (40 converged calculations).

TABLE I. Exp. 5 tetramer results (PBE+D3). [verified: `table1_verification.json`]

System	$E(0\%)$ (Ha)	$E(+5\%)$ (Ha)	α (meV/%)	E_f (eV/dopant)
Pristine	−1363.551	−1363.551	−1.0	—
B-doped	−1328.490	−1328.409	+56.9	+159.0
N-doped	−1414.381	−1414.405	−303.2	−230.5
P-doped	−1370.496	−1370.489	−19.1	−31.5

Observables. α from linear fits to Exp. 5 energies [Table I]. $E_f = (E_{\text{doped}} - E_{\text{pristine}})/n_{\text{dopants}}$ is a relative substitution energy (no μ reference).

We quantify non-additive coupling with the synergy order parameter $\mathcal{S}$:

$$\mathcal{S}(\epsilon, \delta) = \Delta E(\epsilon, \delta) - [\Delta E(\epsilon, 0) + \Delta E(0, \delta)], \quad (1)$$

where ϵ is biaxial strain, $\delta \in \{B, N, P\}$, and $\Delta E(\epsilon, \delta)$ is the energy per atom relative to the pristine, zero-strain reference at the same supercell size $n \times \text{C}_{60}$. Additive superposition gives $\mathcal{S} = 0$; $\mathcal{S} \neq 0$ isolates the cross term. Exp. 10 values are from `c/simukit-sdc (sdc_exp10_results.json)`.

III. RESULTS

A. Dopant-dependent strain sensitivity

Pristine tetramers are strain-insensitive ($|\alpha| \approx 1$ meV/%). Substitution opens a wide spread: N ($\alpha = -303.2$ meV/%), B (+56.9 meV/%), P (−19.1 meV/%; excluding +2.5% metastable point). Figure 1 shows $\Delta E/N_{\text{atom}}$ and α —the central *design* finding is that N and B respond in opposite directions to the same tensile protocol.

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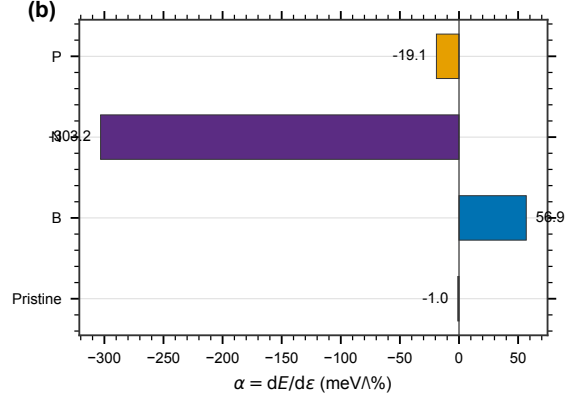
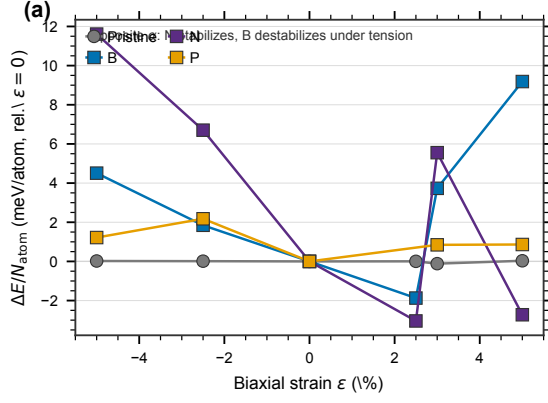


FIG. 1. Exp. 5 strain response. (a) $\Delta E/N_{\text{atom}}$ vs. ϵ . (b) α by dopant; N and B bracket opposite trends under identical loading.

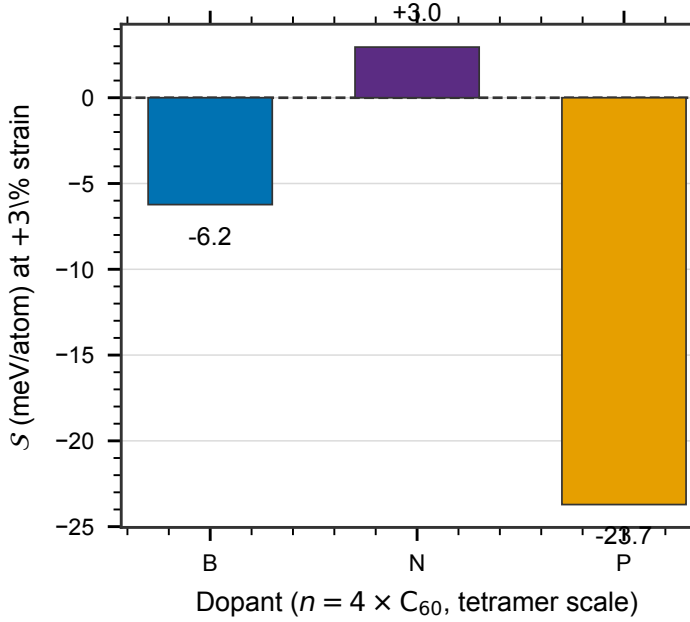


FIG. 2. Synergy S at tetramer scale ($n = 4$, +3% strain). Non-zero bars quantify failure of additive superposition; values from `sdc_exp10_synergy_audit.json`.

B. Quantified non-additivity at tetramer scale

Figure ?? reports S at $n = 4$ and +3% strain [Exp. 10 subset matching tetramer size]. All three dopants deviate from $S = 0$: P shows the largest magnitude ($S \approx -23.7$ meV/atom), B is negative (≈ -6.2 meV/atom), and N is positive ($\approx +3.0$ meV/atom). These residuals are the energy cost of treating strain and substitution as independent—they are directly usable to rank (ϵ, δ) pairs before expensive follow-up calculations.

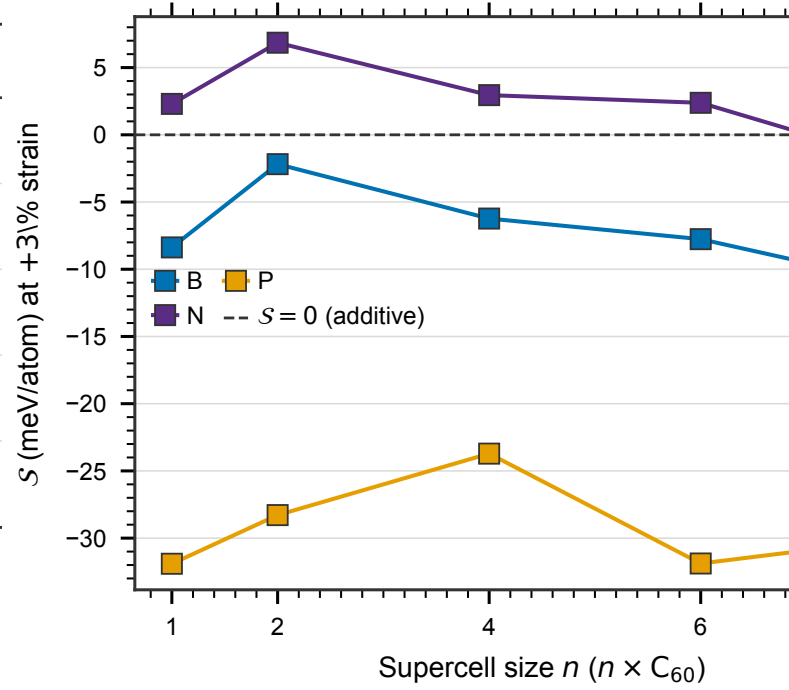


FIG. 3. $S(n)$ at +3% strain (Exp. 10). Dashed line: additive limit. Non-zero values at $n = 8$ support bulk-relevant coupling.

C. Finite-size persistence

$S(n)$ remains non-zero for all B/N/P at every $n = 1-8$ (Fig. 2; 15 synergy points, 40/40 converged). N is positive for $n \leq 6$ and negative at $n = 8$ (-2.3 meV/atom); provisional $S_{\infty} \approx -0.52, -0.68, -0.13$ meV/atom for B, N, P from $1/n$ fits. The cross term therefore does not appear to be a small-cell artefact within the computed range.

IV. DISCUSSION

Design implications. (1) *Do not rank dopants at $\epsilon = 0$ alone:* N is most stable at zero strain but its α implies further stabilization in tension, whereas B becomes less stable under the same load. (2) *Use $\mathcal{S}$ to screen (ϵ, δ) :* at +3% strain and tetramer size, P ($|\mathcal{S}| \approx 24$ meV/atom) shows the strongest non-additive correction; single-knob scans would mis-estimate its energy by orders of magnitude larger than N or B at the same conditions. (3) *Coupled maps beat sequential scans:* Exp. 10 required 40 converged cells but yields a reusable non-additivity metric at each size—cheaper than exhaustive transport or geometry paths for ranking stability trends.

Mechanistically, opposite α signs plausibly reflect π -electron count (N-rich vs. B-hole states) under expanded lattices, but band-resolved and relaxed-ion data are needed for confirmation.

Limitations. Rigid strain; E_f without chemical po-

tentials; N $\mathcal{S}$ sign change by $n = 8$; PBE+D3 only; Exp. 4 two-point audit (IPR/J) does not support a polaron transition claim and is omitted from figures.

V. CONCLUSIONS

qHP C₆₀ graphullerene violates additive strain-doping superposition in total energy. The effect is large (opposite α for N vs. B), quantifiable ($\mathcal{S} \neq 0$ up to ~ 24 meV/atom at tetramer scale), and persistent to $8 \times C_{60}$. For materials design, (ϵ, δ) should be treated as coupled variables; α and $\mathcal{S}$ provide auditable metrics from the DFT workflow archived in this repository.

ACKNOWLEDGMENTS

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